Human Wharton's jelly mesenchymal stem cells: properties, isolation and clinical applications.

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Human Wharton's jelly mesenchymal stem cells (WJ-MSCs) exhibit CD29, CD79 and CD105 markers, characteristic for mesenchymal cell lines. Under the influence of the appropriate factors, WJ-MSCs can be dedifferentiated to osteoblasts, chondrocytes, adipocytes, myocytes, cardiomyocytes, glial cells and dopaminergic neurons. Wharton's jelly (WJ) is one of the potential sources of mesenchymal stem cells (MSCs) - obtaining these cells does not raise moral or ethical objections, because the umbilical cord (UC) is a regular waste material. The expression of the OCT-4 and Nanog proteins, which are characteristic for WJ-MSCs may indicate that these cells have retained some embryonic character. The collected data suggests that WJMSCs show increased division and telomerase activity compared to bone marrow MSCs (BM-MSCs). The published results showed no human leucocyte antigen (HLA) class II expression, with the possibility of HLA class I modification by WJ-MSCs, allowing for the transplantation of these cells both within the same and other species - which allows the use of human cells in animal models. The results of selected studies indicate that WJ-MSCs can be an essential element of regenerative medicine of the 21st century.

A Simple High Yield Technique for Isolation of Wharton's Jelly-derived Mesenchymal Stem Cell.

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The isolation of Mesenchymal Stem Cells (MSCs) from various tissues is possible, with the umbilical cord emerging as a competitive alternative to bone marrow. In order to fulfill the demands of cell therapy, it is essential to generate stem cells on a clinical scale while minimizing time, cost, and contamination. Here is a simple and effective protocol for isolating MSC from Wharton's Jelly (WJ-MSC) using the explant method with various supplements. Utilizing the explant method, small fragments of Wharton's jelly from the human umbilical cord were cultured in a flask. The multipotency of the isolated cells, were confirmed by their differentiation ability to osteocyte and adipocyte. Additionally, the immunophenotyping of WJ-MSCs showed positive expression of CD73, CD90, and CD105, while remaining negative for hematopoietic markers CD34 and CD45, meeting the criteria for WJ-MSC identification. Following that, to evaluate cells' proliferative capacity, various supplements, including basic Fibroblast Growth Factor (bFGF), Non-Essential amino acids (NEA), and L-Glutamine (L-Gln) were added to either alpha-Minimal Essential Medium (α-MEM) or Dulbecco's Modified Eagle's Medium-F12 (DMEM-F12), as the basic culture media. WJ-MSCs isolated by the explant method were removed from the tissue after seven days and transferred to the culture medium. These cells differentiated into adipocyte and osteocyte lineages, expressing CD73, CD90, and CD105 positively and CD34 and CD45 negatively. The results revealed that addition of bFGF to α-MEM or DMEMF12 media significantly increased the proliferation of MSCs when compared to the control group. However, there were no significant differences observed when NEA or LGln were added. Although bFGF considerably enhances cell proliferation, our study demonstrates that MSCs can grow and expand when properly prepared Wharton's jelly tissues of the human umbilical cord.

Regenerative potential of Wharton's jelly-derived mesenchymal stem cells: A new

horizon of stem cell therapy.

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Umbilical cord Wharton's jelly-derived mesenchymal stem cells (WJ-MSCs) have recently gained considerable attention in the field of regenerative medicine. Their high proliferation rate, differentiation ability into various cell lineages, easy collection procedure, immuno-privileged status, nontumorigenic properties along with minor ethical issues make them an ideal approach for tissue repair. Besides, the number of WJ-MSCs in the umbilical cord samples is high as compared to other sources. Because of these properties, WJ-MSCs have rapidly advanced into clinical trials for the treatment of a wide range of disorders. Therefore, this paper summarized the current preclinical and clinical studies performed to investigate the regenerative potential of WJ-MSCs in neural, myocardial, skin, liver, kidney, cartilage, bone, muscle, and other tissue injuries.

Human Wharton's Jelly-Derived Stem Cells Display a Distinct Immunomodulatory and Proregenerative Transcriptional Signature Compared to Bone Marrow-Derived Stem Cells.

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Mesenchymal stromal cells (MSCs) are multipotent stem cells with immunosuppressive and trophic support functions. While MSCs from different sources frequently display a similar appearance in culture, they often show differences in their surface marker and gene expression profiles. Although bone marrow is considered the "gold standard" tissue to isolate classical MSCs (BM-MSC), MSC-like cells are currently also derived from more easily accessible extra-embryonic tissues such as the umbilical cord. In this study, we defined the best way to isolate MSCs from the Wharton's jelly of the human umbilical cord (WJ-MSC) and assessed the mesenchymal and immunological phenotype of BM-MSC and WJ-MSC. Moreover, the gene expression profile of established WJ-MSC cultures was compared to two different bone marrow-derived stem cell populations (BM-MSC and multipotent adult progenitor cells or MAPC®). We observed that explant culturing of Wharton's jelly matrix is superior to collagenase tissue digestion for obtaining mesenchymal-like cells, with explant isolated cells displaying increased expansion potential. While being phenotypically similar to adult MSCs, WJ-MSC show a different gene expression profile. Gene ontology analysis revealed that genes associated with cell adhesion, proliferation, and immune system functioning are enriched in WJ-MSC. In vivo transplantation confirms their immune modulatory effect on T cells, similar to BM-MSC and MAPC. Furthermore, WJ-MSC intrinsically overexpress genes involved in neurotrophic support and their secretome induces neuronal maturation of SH-SY5Y neuroblastoma cells to a greater extent than BM-MSC. This signature makes WJ-MSC an attractive candidate for cell-based therapy in neurodegenerative and immune-mediated central nervous system disorders such as multiple sclerosis, Parkinson's disease, or amyotrophic lateral sclerosis.

The Molecular Regulatory Mechanism in Multipotency and Differentiation of Wharton's Jelly Stem Cells.

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Wharton's jelly-derived mesenchymal stem cells (WJ-MSCs) are isolated from Wharton's jelly tissue of umbilical cords. They possess the ability to differentiate into lineage cells of three germ layers. WJ-MSCs have robust proliferative ability and strong immune modulation capacity. They can be easily collected and there are no ethical problems associated with their use. Therefore, WJ-MSCs have great tissue engineering value and clinical application prospects. The identity and functions of WJ-MSCs are regulated by multiple interrelated regulatory mechanisms, including transcriptional regulation and epigenetic modifications. In this article, we summarize the latest research progress on the genetic/epigenetic regulation mechanisms and essential signaling pathways that play crucial roles in pluripotency and differentiation of WJ-MSCs.

Method to isolate mesenchymal-like cells from Wharton's Jelly of umbilical cord.

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The umbilical cord is a noncontroversial source of mesenchymal-like stem cells. Mesenchymal-like cells are found in several tissue compartments of the umbilical cord, placenta, and decidua. Here, we confine ourselves to discussing mesenchymal-like cells derived from Wharton's Jelly, called umbilical cord matrix stem cells (UCMSCs). Work from several laboratories shows that these cells have therapeutic potential, possibly as a substitute cell for bone marrow-derived mesenchymal stem cells for cellular therapy. There have been no head-to-head comparisons between mesenchymal cells derived from different sources for therapy; therefore, their relative utility is not understood. In this chapter, the isolation protocols of the Wharton's Jelly-derived mesenchymal cells are provided as are protocols for their in vitro culturing and storage. The cell culture methods provided will enable basic scientific research on the UCMSCs. Our vision is that both umbilical cord blood and UCMSCs will be commercially collected and stored in the future for preclinical work, public and private banking services, etc. While umbilical cord blood banking standard operating procedures exist, the scenario mentioned above requires clinical-grade UCMSCs. The hurdles that have been identified for the generation of clinical-grade umbilical cord-derived mesenchymal cells are discussed.

3D Decellularized Native Extracellular Matrix Scaffold for In Vitro Culture Expansion of Human Wharton's Jelly-Derived Mesenchymal Stem Cells (hWJ MSCs).

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Mesenchymal stem cells (MSCs) are derived from Wharton's jelly tissue of the human umbilical cord. Given appropriate culture conditions, these cells can self-renew and differentiate into multiple cell types across the lineages. Among the properties exhibited by these cells, immunomodulation through secretion of trophic factors has been widely exploited in a broad spectrum of preclinical/clinical regenerative applications. Moreover, the extracellular matrix is found to play a major role apart from niche cells in determining stem cell fate including that of MSCs. Therefore, the currently employed technique of two-dimensional culture expansion can alter the inherent properties of naïve MSCs originally residing within the three-dimensional space. This limitation can be overcome to some extent by using native extracellular matrix scaffold culture system which mimics the in situ microenvironment. In this chapter, we have elucidated the protocol for the preparation of a native extracellular matrix scaffold by decellularization of the MSC sheet and thereof culture expansion and characterization of human Wharton's jelly-derived MSCs.

Human Wharton's Jelly Mesenchymal Stem Cells plasticity augments scar-free skin wound healing with hair growth.

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Human mesenchymal stem cells (MSCs) are a promising candidate for cell-based transplantation and regenerative medicine therapies. Thus in the present study Wharton's Jelly Mesenchymal Stem Cells (WJ-MSCs) have been derived from extra embryonic umbilical cord matrix following removal of both arteries and vein. Also, to overcome the clinical limitations posed by fetal bovine serum (FBS) supplementation because of xenogeneic origin of FBS, usual FBS cell culture supplement has been replaced with human platelet lysate (HPL). Apart from general characteristic features of bone marrow-derived MSCs, wharton jelly-derived MSCs have the ability to maintain phenotypic attributes, cell growth kinetics, cell cycle pattern, in vitro multilineage differentiation plasticity, apoptotic pattern, normal karyotype-like intrinsic mesenchymal stem cell properties in long-term in vitro cultures. Moreover, the WJ-MSCs exhibited the in vitro multilineage differentiation capacity by giving rise to differentiated cells of not only mesodermal lineage but also to the cells of ectodermal and endodermal lineage. Also, WJ-MSC did not present any aberrant cell state upon in vivo transplantation in SCID mice and in vitro soft agar assays. The immunomodulatory potential assessed by gene expression levels of immunomodulatory factors upon exposure to inflammatory cytokines in the fetal WJ-MSCs was relatively higher compared to adult bone marrow-derived MSCs. WJ-MSCs seeded on decellularized amniotic membrane scaffold transplantation on the skin injury of SCID mice model demonstrates that combination of WJ-MSCs and decellularized amniotic membrane scaffold exhibited significantly better wound-healing capabilities, having reduced scar formation with hair growth and improved biomechanical properties of regenerated skin compared to WJ-MSCs alone. Further, our experimental data indicate that indocyanin green (ICG) at optimal concentration can be resourcefully used for labeling of stem cells and in vivo tracking by near infrared fluorescence non-invasive live cell imaging of labelled transplanted cells, thus proving its utility for therapeutic applications.

Mesenchymal stem cells derived from Wharton's jelly: comparative phenotype analysis between tissue and in vitro expansion.

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Mesenchymal stem cells (MSCs) are useful multipotent stem cells that are found in many tissues. While MSCs can usually be isolated from adults via bone marrow aspiration (BM-MSCs), MSCs derived from the discarded umbilical cord, more precisely from Wharton's jelly (WJ), offer a low-cost and pain-free collection method of MSCs that may be cryogenically stored, and are considered extremely favorable for tissue engineering purpose. The aim of this study was to analyze the harvested number of cells per centimeter of human umbilical cord (UC) and carry out the phenotype of these WJ-MSCs after explant or enzymatic methods. Fresh UCs were obtained from full-term births, and processed within 6 hours from partum to obtain the WJ-MSCs. UC sections were analyzed in confocal microscopy to analyze cells phenotype in situ. Others UC components were treated either by enzymatic method or by explant method to obtain isolated cells and to analyze cells phenotype until the end of the first passage. We have successfully generated MSCs from UC by using explant and enzymatic methods. Using microscopy confocal, we identified the expression of some MSCs markers in situ of Wharton's jelly tissue as well as in perivascular region. Our comparative study, between explant and enzymatic digestion, indicated, that WJ expressed most of MSCs markers in both conditions, but a remarkable variation of cell phenotype expression was distinguished after primary culture comparing to directly isolated cells by enzymatic digestion. We also studied the expression of CD271, which showed to be weakly expressed in situ on fresh fragment of WJ.

Effects of intravascular administration of mesenchymal stromal cells derived from Wharton's Jelly of the umbilical cord on systemic immunomodulation and neuroinflammation after traumatic brain injury (TRAUMACELL): study protocol for a multicentre randomised controlled trial.

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Traumatic brain injury (TBI) is one of the leading causes of death and disability worldwide. Treatments for TBI patients are limited and none has been shown to provide prolonged and long-term neuroprotective or neurorestorative effects. A growing body of evidence suggests a link between TBI-induced neuro-inflammation and neurodegenerative post-traumatic disorders. Consequently, new therapies triggering immunomodulation and promoting neurological recovery are the subject of major research efforts. We hypothesise that repeated intravenous treatment with mesenchymal stromal cells derived from Wharton's Jelly of the umbilical cord-derived mesenchymal stromal cells ((WJ-UC-MSC) may be associated with a significant decrease of post-TBI neuroinflammation and improvement of neurological status. The TRAUMACELL trial is a prospective, national multicentre, phase III, superiority, double-arm comparative randomised (1:1) double-blinded clinical trial. Among patients aged between 18-50, with a severe TBI defined by a Glasgow score less than 12 (within the first 48 hours) with brain traumatic lesion on CT Scan and needing intracranial pressure monitoring, with no other significant organ trauma (abbreviated injury scale<2) and unresponsive to verbal commands after 5 days of sedation discontinuation, 68 will be randomly allocated to receive either WJ-UC-MSC solution or placebo, with three intravenous injections 1 week apart. The primary outcome is the [18F]-DPA-714 signal intensity in corpus callosum measured by dynamic positron emission tomography (PET)-MRI at 6 months after the last injection, blinded to the randomisation arm, to evaluate the post-traumatic neuro-inflammation. The TRAUMACELL trial has been approved by an independent ethics committee (CPP SUD EST II) and French Medicines Agency (2023-504415-33-00) for all study centres. Participant recruitment will be starting in September 2024. Results will be published in international peer-reviewed medical journals. NCT06146062, first posted 24 November 2023 PROTOCOL VERSION IDENTIFIER: TRAUMACELL-V.2.0_20240102.